

In [3]:

```
def print_ten_numbers():  
    for i in range(1, 11):  
        print(i)  
  
print_ten_numbers()
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

In [4]:

```
def sum_n_numbers(n):  
    return (n * (n + 1)) // 2  
  
n = int(input("Enter N: "))  
print("Sum:", sum_n_numbers(n))
```

```
Enter N: 6  
Sum: 21
```

In [5]:

```
def reverse_number(num):  
    rev = 0  
    while num > 0:  
        rev = (rev * 10) + (num % 10)  
        num //= 10  
    return rev  
  
number = int(input("Enter a number to reverse: "))  
print("Reversed:", reverse_number(number))
```

```
Enter a number to reverse: 34  
Reversed: 43
```

In [6]:

```
def count_digits(num):  
    if num == 0:  
        return 1  
    count = 0  
    while num > 0:  
        count += 1  
        num //= 10  
    return count  
  
number = int(input("Enter a number to count digits: "))  
print("Total digits:", count_digits(number))
```

Enter a number to count digits: 2345
Total digits: 4

In [7]:

```
def is_palindrome(num):
    # Reuses the logic from the previous reverse function
    temp = num
    rev = 0
    while temp > 0:
        rev = (rev * 10) + (temp % 10)
        temp //= 10
    return num == rev

number = int(input("Enter a number to check palindrome: "))
print("Is Palindrome?:", is_palindrome(number))
```

Enter a number to check palindrome: 45
Is Palindrome?: False

In [8]:

```
def generate_fibonacci(n):
    series = []
    a, b = 0, 1
    for _ in range(n):
        series.append(a)
        a, b = b, a + b
    return series

terms = int(input("Enter number of Fibonacci terms: "))
print("Series:", generate_fibonacci(terms))
```

Enter number of Fibonacci terms: 9
Series: [0, 1, 1, 2, 3, 5, 8, 13, 21]

In [9]:

```
def add(a, b): return a + b
def subtract(a, b): return a - b
def multiply(a, b): return a * b
def divide(a, b): return a / b if b != 0 else "Error: Division by zero"

def calculator():
    print("1. Add\n2. Subtract\n3. Multiply\n4. Divide")
    choice = input("Select operation (1-4): ")
    if choice in ['1', '2', '3', '4']:
        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))

        if choice == '1': print("Result:", add(num1, num2))
        elif choice == '2': print("Result:", subtract(num1, num2))
        elif choice == '3': print("Result:", multiply(num1, num2))
        elif choice == '4': print("Result:", divide(num1, num2))
    else:
        print("Invalid Input")
```

```
calculator()
```

```
1. Add
2. Subtract
3. Multiply
4. Divide
Select operation (1-4): 3
Enter first number: 34
Enter second number: 6
Result: 204.0
```

```
In [10]: def save_student_details(filename, name, roll_no, marks):
        with open(filename, 'w') as f:
            f.write(f"Name: {name}\nRoll No: {roll_no}\nMarks: {marks}\n")

        save_student_details("student.txt", "Alice", "ECE-01", 95)
        print("File written successfully.")
```

File written successfully.

```
In [11]: def read_file_data(filename):
        try:
            with open(filename, 'r') as f:
                print(f.read())
        except FileNotFoundError:
            print("File not found.")

        read_file_data("student.txt")
```

Name: Alice
Roll No: ECE-01
Marks: 95

```
In [12]: def safe_division(numerator, denominator):
        try:
            result = numerator / denominator
            return result
        except ZeroDivisionError:
            return "Division by zero error handled smoothly."

        num = float(input("Enter numerator: "))
        den = float(input("Enter denominator: "))
        print("Result:", safe_division(num, den))
```

Enter numerator: 5
Enter denominator: 8
Result: 0.625

```
In [13]: class Student:
        def __init__(self, name, marks):
            self.name = name
            self.marks = marks
```

```
        def display(self):
            print(f"Student Name: {self.name}, Marks: {self.marks}")

# Testing the class
s1 = Student("Ananya", 92)
s1.display()
```

Student Name: Ananya, Marks: 92